Lab 2 - Gray Level Slicing

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1 Theory

Gray level slicing is a technique used in image processing to highlight specific ranges of intensity values within an image. This can be achieved in two main ways: - By setting all pixel values within a specified range to a single value (e.g., white) and all others to another value (e.g., black), resulting in a binary image. - By enhancing (brightening or darkening) the pixel values within the range of interest while leaving the rest of the image unchanged.

This method is particularly useful for emphasizing features in an image that fall within a certain intensity range, making them more visible for analysis.

[19]:
```python
#import libraries
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt
```

[20]:
```python
#read the image
# reading image and converting to gray scale
img = Image.open('img/samir.jpg').convert('L')
# display image
img
```

[20]:

[21]:
```
img = img.resize((400,400), Image.Resampling.LANCZOS)
# convert to numpy array
numpy_image = np.array(img)
```

[22]:
```
numpy_image.shape
row = numpy_image.shape[0]
column = numpy_image.shape[1]
```

- Set all the pixel values with a range of interest to one value (white) and all others to another value (black) produces a binary image.

[23]:
```
new_array = np.zeros(shape=(row,column))
```

[24]:
```python
for i in range(row):
  for j in range(column):
    new_array[i][j] = numpy_image[i][j]
```

[25]:
```python
new_array
```

[25]:
```
array([[180., 180., 178., …, 151., 151., 151.],
       [180., 180., 179., …, 151., 150., 151.],
       [180., 181., 180., …, 152., 151., 152.],
       …,
       [ 39.,  35.,  34., …, 133., 134., 116.],
       [ 36.,  33.,  35., …, 134., 132., 131.],
       [ 35.,  32.,  34., …, 135., 132., 133.]])
```

[26]:
```python
numpy_image
```

[26]:
```
array([[180, 180, 178, …, 151, 151, 151],
       [180, 180, 179, …, 151, 150, 151],
       [180, 181, 180, …, 152, 151, 152],
       …,
       [ 39,  35,  34, …, 133, 134, 116],
       [ 36,  33,  35, …, 134, 132, 131],
       [ 35,  32,  34, …, 135, 132, 133]], dtype=uint8)
```

[27]:
```python
for i in range(row):
  for j in range(column):
    if((numpy_image[i][j]>100)&(numpy_image[i][j]<150)):
      new_array[i][j] = 255
    else:
      new_array[i][j] = 0
```

[28]:
```python
new_array
```

[28]:
```
array([[  0.,   0.,   0., …,   0.,   0.,   0.],
       [  0.,   0.,   0., …,   0.,   0.,   0.],
       [  0.,   0.,   0., …,   0.,   0.,   0.],
       …,
       [  0.,   0.,   0., …, 255., 255., 255.],
       [  0.,   0.,   0., …, 255., 255., 255.],
       [  0.,   0.,   0., …, 255., 255., 255.]])
```

[29]:
```python
gray_level_slicing_image = Image.fromarray(new_array)
gray_level_slicing_image = gray_level_slicing_image.convert("L")
gray_level_slicing_image
```

[29]:

- Brighten(or darken) pixel values in a range of interest and leave all others unchanged.

[30]:
```
new_array2 = np.zeros(shape=(row,column))
```

[31]:
```
for i in range(row):
  for j in range(column):
    if((numpy_image[i][j]>100)&(numpy_image[i][j]<150)):
      new_array2[i][j] = 0
    else:
      new_array2[i][j] = numpy_image[i][j]
```

[32]:
```
gray_level_slicing_image2 = Image.fromarray(new_array2)
gray_level_slicing_image2 = gray_level_slicing_image2.convert("L")
gray_level_slicing_image2
```

[32]:

2 Observation

- The original grayscale image was loaded and resized for processing.
- A binary image was created by setting pixels with intensity values between 100 and 150 to white (255) and all others to black (0). This clearly highlighted the regions of the image within the specified intensity range.
- In the second approach, pixels within the range of 100 to 150 were set to black (0), while all other pixels retained their original intensity values. This allowed the features outside the specified range to remain visible, while the range of interest was suppressed.

3 Conclusion

Gray level slicing effectively highlights or suppresses specific intensity ranges in an image. The binary approach is useful for isolating features, while the enhancement approach allows for selective emphasis or suppression of certain regions. This technique is valuable in applications where particular intensity ranges correspond to important features, such as medical imaging or industrial inspection.